

Lecture 7: Introduction to Contextual Bandits

ID 6002W: Online and Reinforcement Learning

Web-enabled M.Tech. program, IIT Madras

The Story So Far

Multi-armed
Bandit

Best-arm
Identification

Regret
Minimisation

UCB

Thompson
Sampling

estimate → quantify uncertainty → act using optimism or sampling

Assumptions so far: fixed, featureless arms

one user type · fixed arm set · fixed means $\mu_1, \dots, \mu_K$

How realistic are these assumptions?

Yahoo! Front Page, 2009

A real personalisation problem

- ▶ Setting: Four editor-curated articles shown at a time on yahoo.com (F1–F4, refreshed hourly)
- ▶ Problem statement: choose one article of these four as main story, based on individual interests
- ▶ Reward signal: whether the user clicks the highlighted story



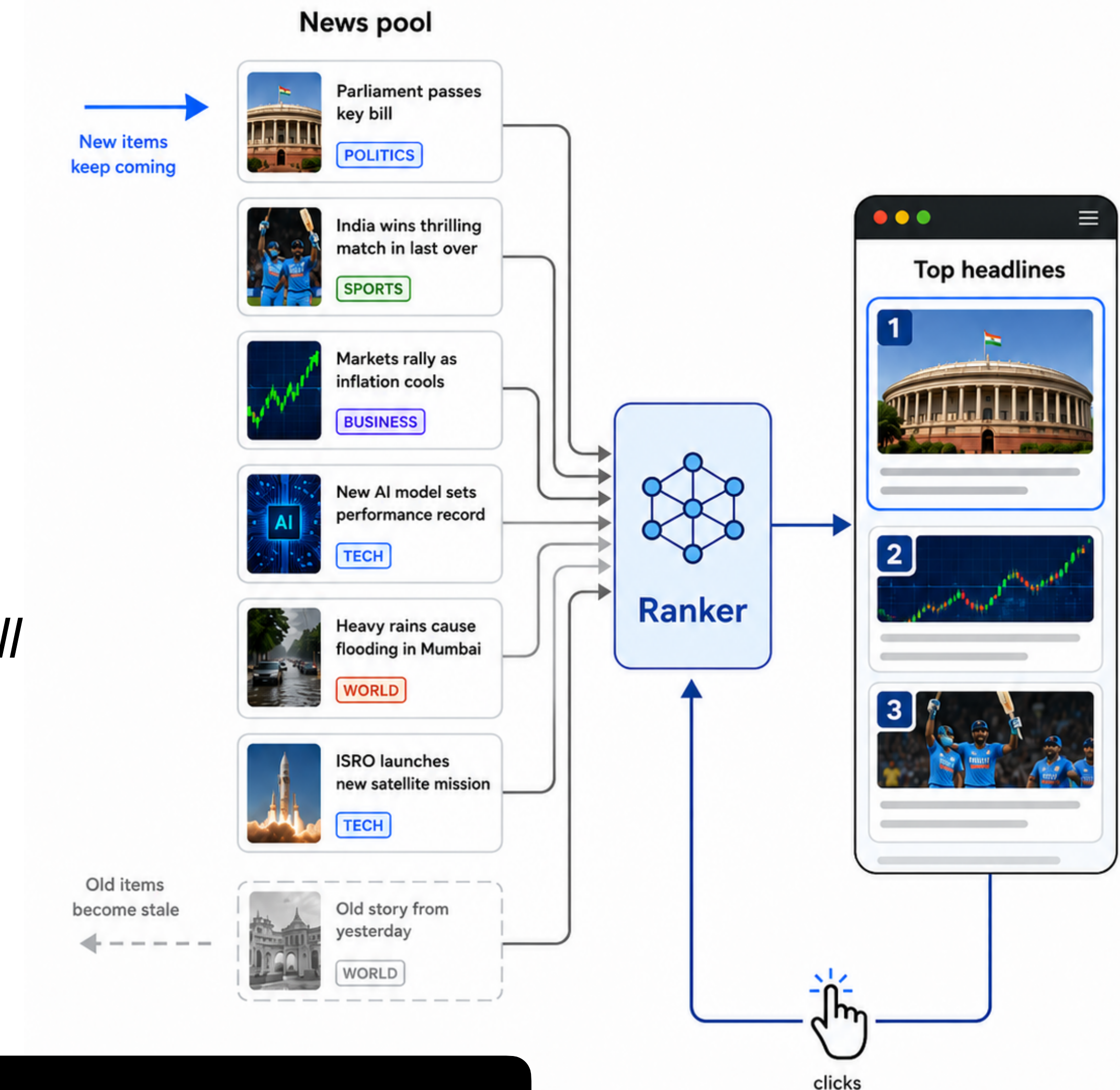
Figure 1: A snapshot of the “Featured” tab in the Today Module on Yahoo! Front Page. By default, the article at F1 position is highlighted at the story position.

Influential paper: A Contextual-Bandit Approach to Personalised News Article Recommendation

Where The Assumptions Break

In real-world systems

- **Arms change**
 - *New items arrive, old items go away*
- **Users differ**
 - *We need personalisation; no single best item for all*
- **Many arms**
 - *Can't explore all options one by one*



Learning about one user/item should help with similar users/items

Describing Items By Features

Why features buy you transfer

- In vanilla bandits, an arm is just an index i
- Now, each arm a comes with feature vector x_a
 - Example: a news article may have features like topic, length, recency, source
- Similar features $\Rightarrow$ similar predictions
 - Learning about one article gives some idea about other similar articles — even a new one

Old view



μ_A



μ_B



μ_C

Each arm learned separately

A new article may be unseen, but its features are not!

Features Have Semantic Meaning



The Contextual Bandit Protocol

- At each round $t = 1, 2, \dots$
 - Observe user u_t and an arm set $\mathcal{A}_t$, with a feature vector $x_{t,a}$ for each arm a
 - Choose an arm $a_t \in \mathcal{A}_t$
 - Observe reward r_{t,a_t} — only for the chosen arm
 - Update

NEW: a feature vector
per arm, per round

SAME: bandit feedback —
we only get pulled arm's reward

What goes into the feature $x_{t,a}$?

What is the quantity to be learned?

From Counting to Predicting

- Until now, learning meant estimating an arm reward by simple averaging:

pull arm i many times, average the rewards $\rightarrow \hat{\mu}_i$

- With features, estimating an arm becomes a prediction problem: $x_a \xrightarrow{f} \hat{\mu}_a$

Does this remind you of some supervised learning problem?

- When rewards are binary $\rightarrow$ classification
- When rewards are real $\rightarrow$ regression

Contextual Bandits with Item Features

Basic Modelling Assumptions

We need to learn a function from features to rewards

- The simplest possible function is a *linear function*
- We model the expected reward as a dot product:

$$\mu_a = \mathbb{E}[X_t | A_t = x_a] = \langle \theta, x_a \rangle$$

Exactly like
linear regression

- One *unknown* weight vector θ , shared across all arms
- All arm features $x_a \in \mathbb{R}^d \Rightarrow \theta \in \mathbb{R}^d$

θ is the parameter to be learned (estimated) from data

Contextual Bandits with Item Features

Basic Modelling Assumptions

We need to learn a function from features to rewards

- Linear functions model real-valued rewards. What if the reward is binary?
- We pass the linear model through a non-linear link function

$$\mu_a = \sigma \left(\langle \theta, x_a \rangle \right)$$

Exactly like
logistic regression

$$\sigma(t) = 1/(1 + \exp(-t))$$

- Once again, we have one *unknown* weight vector θ , shared across all arms
- Similar to linear model. We will focus on linear model in this course

Contextual Bandits with Item Features

What we gain, what we miss

- **The gain:**

- a brand-new article (published five minutes ago) has *known* features x
- From past data, we have an *estimate* $\hat{\theta}$
- $\rightarrow \langle \hat{\theta}, x \rangle$ gives an immediate estimate of article quality, before a single click

Arm churn: ✓

- **The miss:**

- $\langle \hat{\theta}, x \rangle$ depends only on the article, not on who is asking
- A sports fan and a finance professional get identical recommendations

Personalisation: ✗

Can incorporating user features give us personalisation?

Personalisation

First attempt

Can we learn a separate contextual bandit for each user?

Replace $\theta \rightarrow \theta_u$ to get $\mu_{u,a} = \langle \theta_u, x_a \rangle$

Does not work well: too little data per user!

What if we split user into groups?

Sports fans $\rightarrow \theta_{\text{sports}}$

Investors $\rightarrow \theta_{\text{finance}}$

Cinema buffs $\rightarrow \theta_{\text{cinema}}$

Better, but brittle. How many groups? How to sort a new user?

Best of Both Worlds

Combining user and item features

$$x_{t,a} = \phi(\text{user at time } t, \text{article } a)$$

Learn one shared θ across all users and items:

$$\mu_{t,a} = \mathbb{E}[X_{t,a} \mid x_{t,a}] = \langle \theta, x_{t,a} \rangle$$

✓ Personalised: $x_{t,a}$ depends on the user

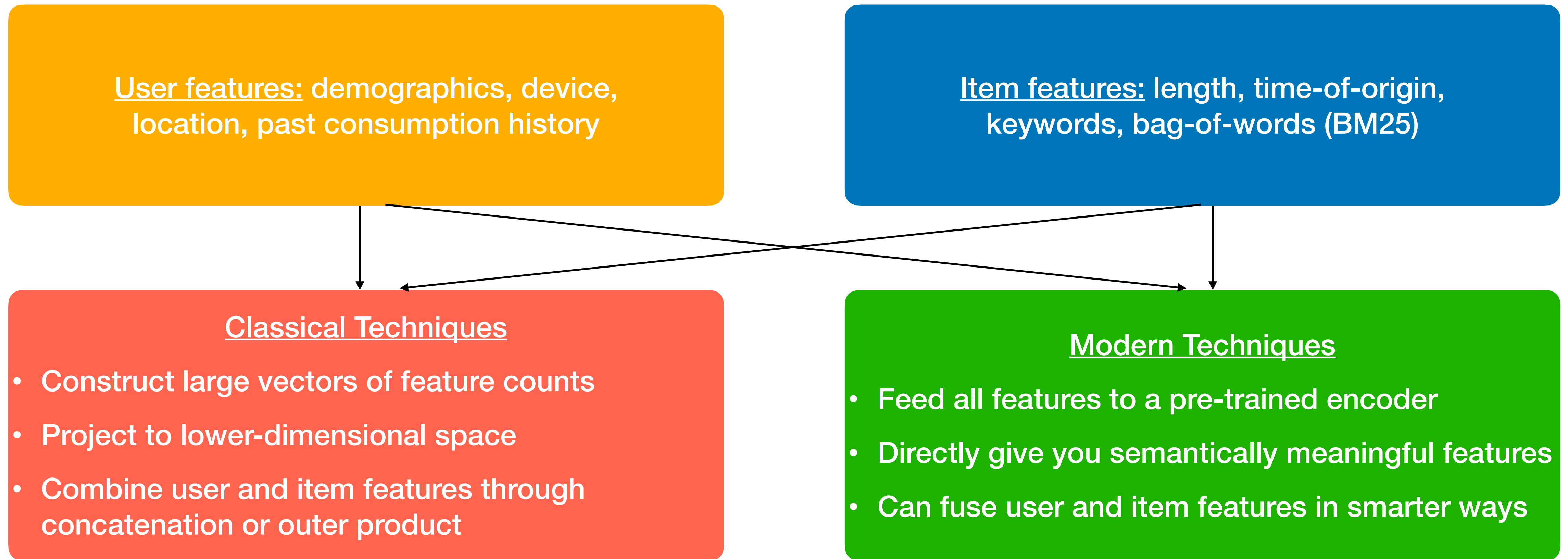
✓ Knowledge transfer: θ is learned across users and items

✓ Flexible: incorporates known aspects of new users and items

Goal: choose arms with high reward $\rightarrow$ estimate θ

How Do We Get Features?

An engineering challenge



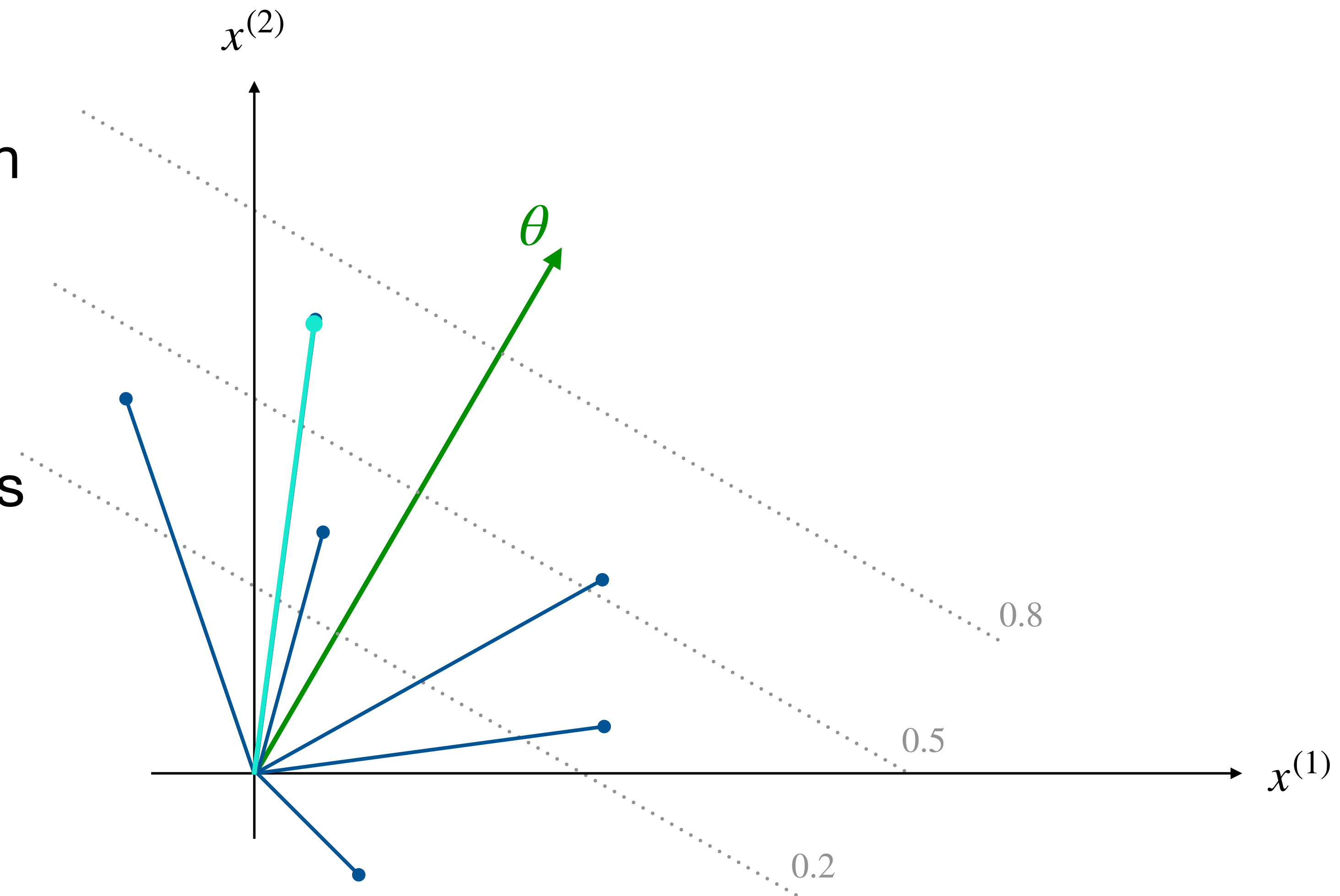
We will cover more practical aspects in weeks 6/7

A Deeper Dive Into Linear Bandits

Arms in Feature Space

Geometry of the model

- Let the set of arms $\mathcal{A}_t$ be given
- It can be represented by a set of points in $\mathbb{R}^d$
- The unknown vector θ provides a *direction of preference*
- Good arms are those that are better aligned to θ
- Finding good arms reduces to estimating θ



How Do We Find θ ?

Least squares estimation

After n rounds, we have observed $(x_1, r_1), (x_2, r_2), \dots, (x_n, r_n)$

Where $x_s = x_{s,a_s}$ is the feature vector of the arm actually chosen

Estimate θ by solving the least-squares problem

$$\hat{\theta} = \arg \min_{\theta} \sum_{s=1}^n \left(r_s - \langle \theta, x_s \rangle \right)^2$$

Difference from linear regression: you can *control* the features x_s that is used in the estimate

Importance of Regulariser

Ridge regression

In bandits, early data can be scarce $\Rightarrow$ least squares problem does not have a solution

Solution: add a regulariser to the least-squares formulation

$$\hat{\theta} = \arg \min_{\theta} \sum_{s=1}^n (r_s - \langle \theta, x_s \rangle)^2 + \lambda \|\theta\|^2$$

Equivalent closed form:

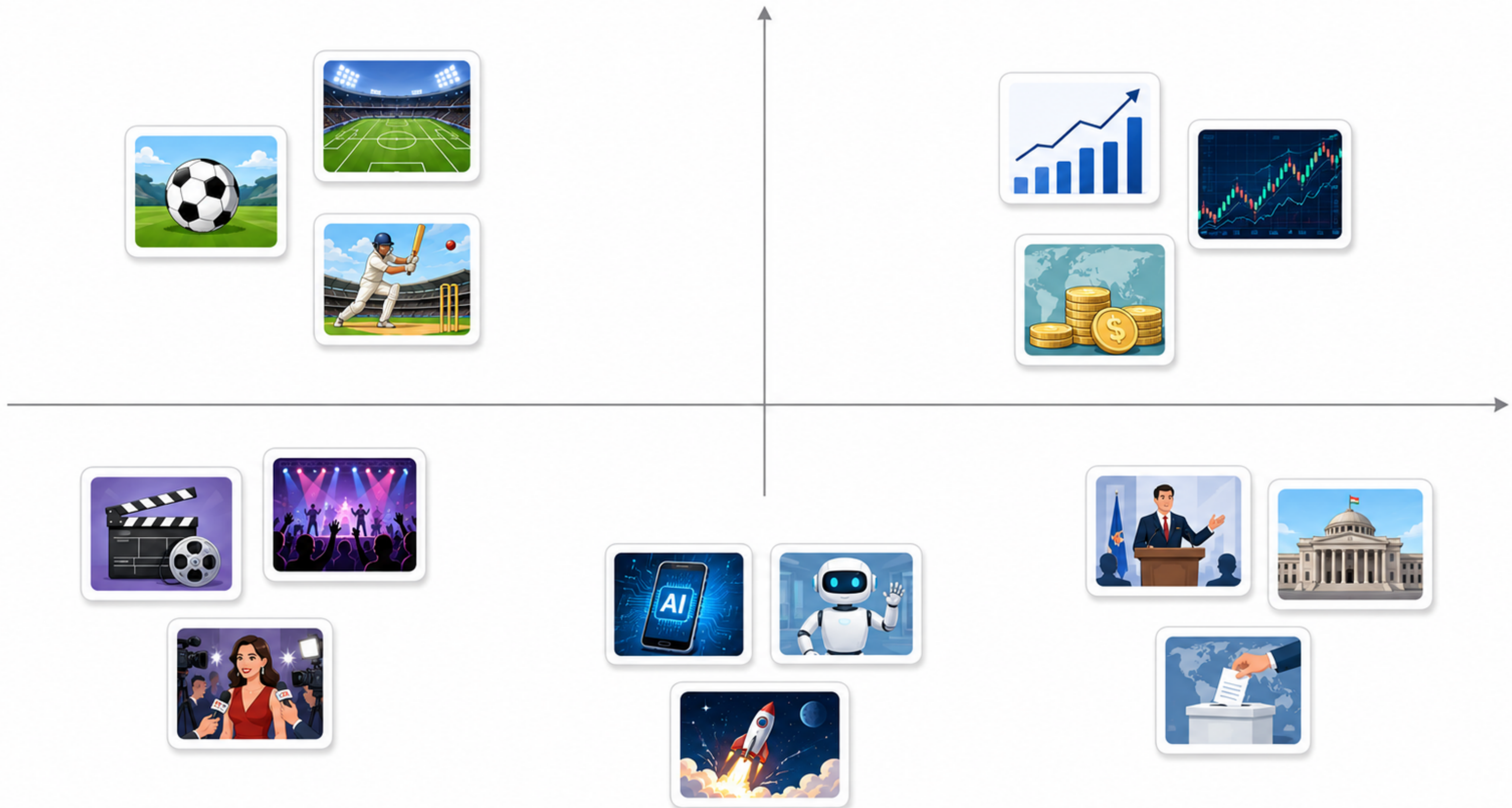
$$\hat{\theta} = V_n^{-1} b_n$$

$$V_n = \lambda I + \sum_{s=1}^n x_s x_s^T$$

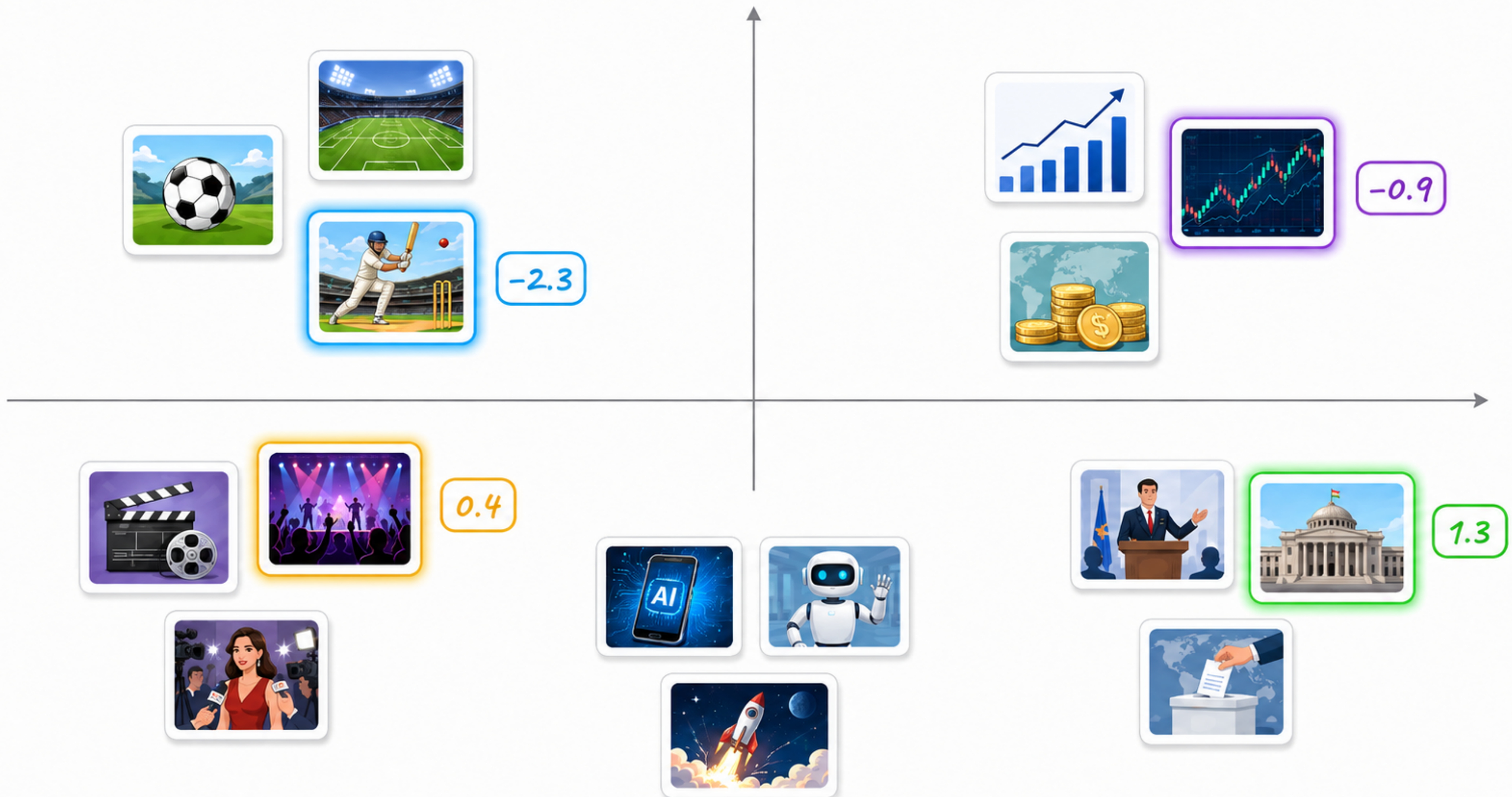
$$b_n = \sum_{s=1}^n x_s r_s$$

This matrix will be very important going forward

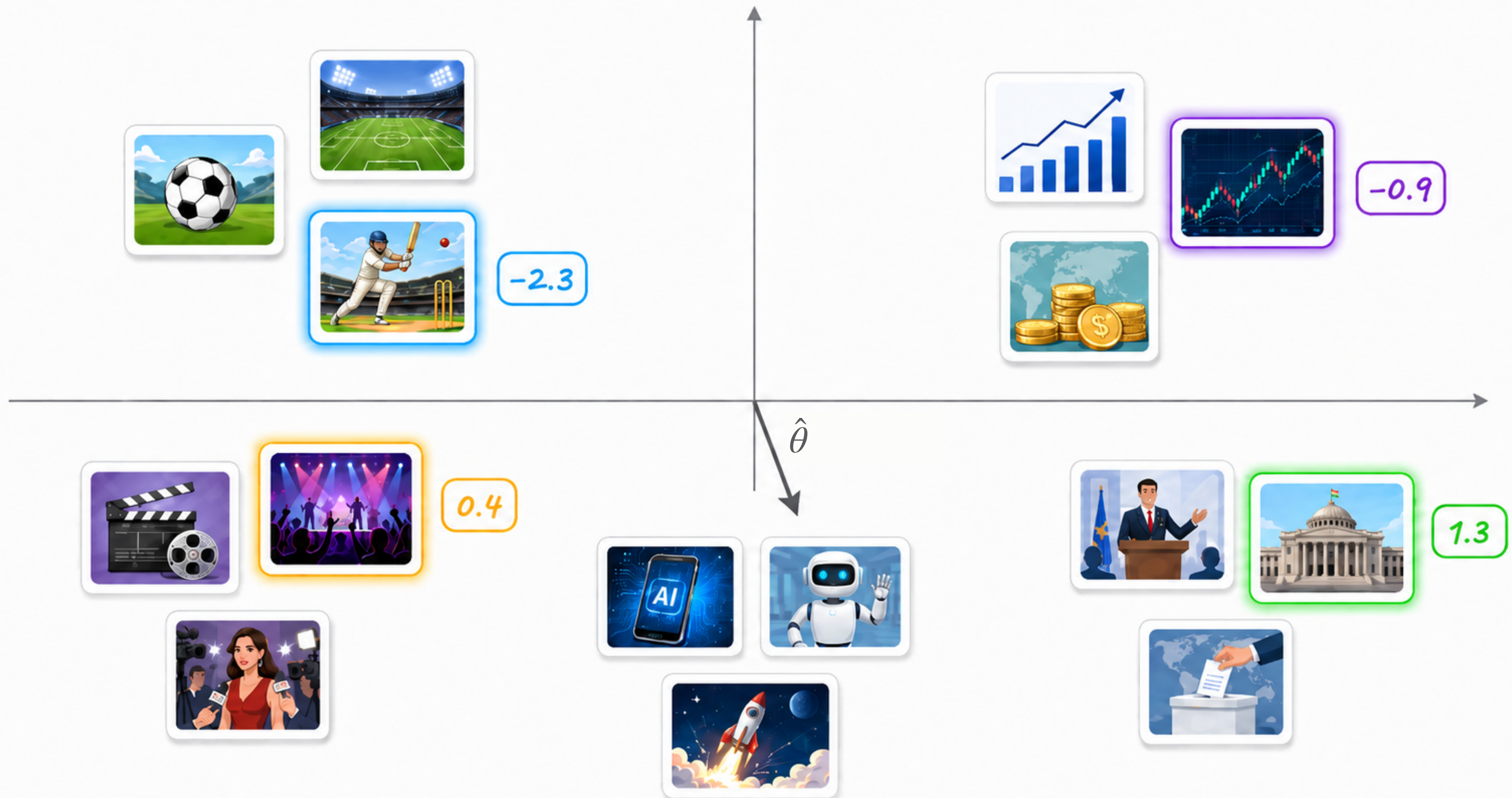
A Worked Example



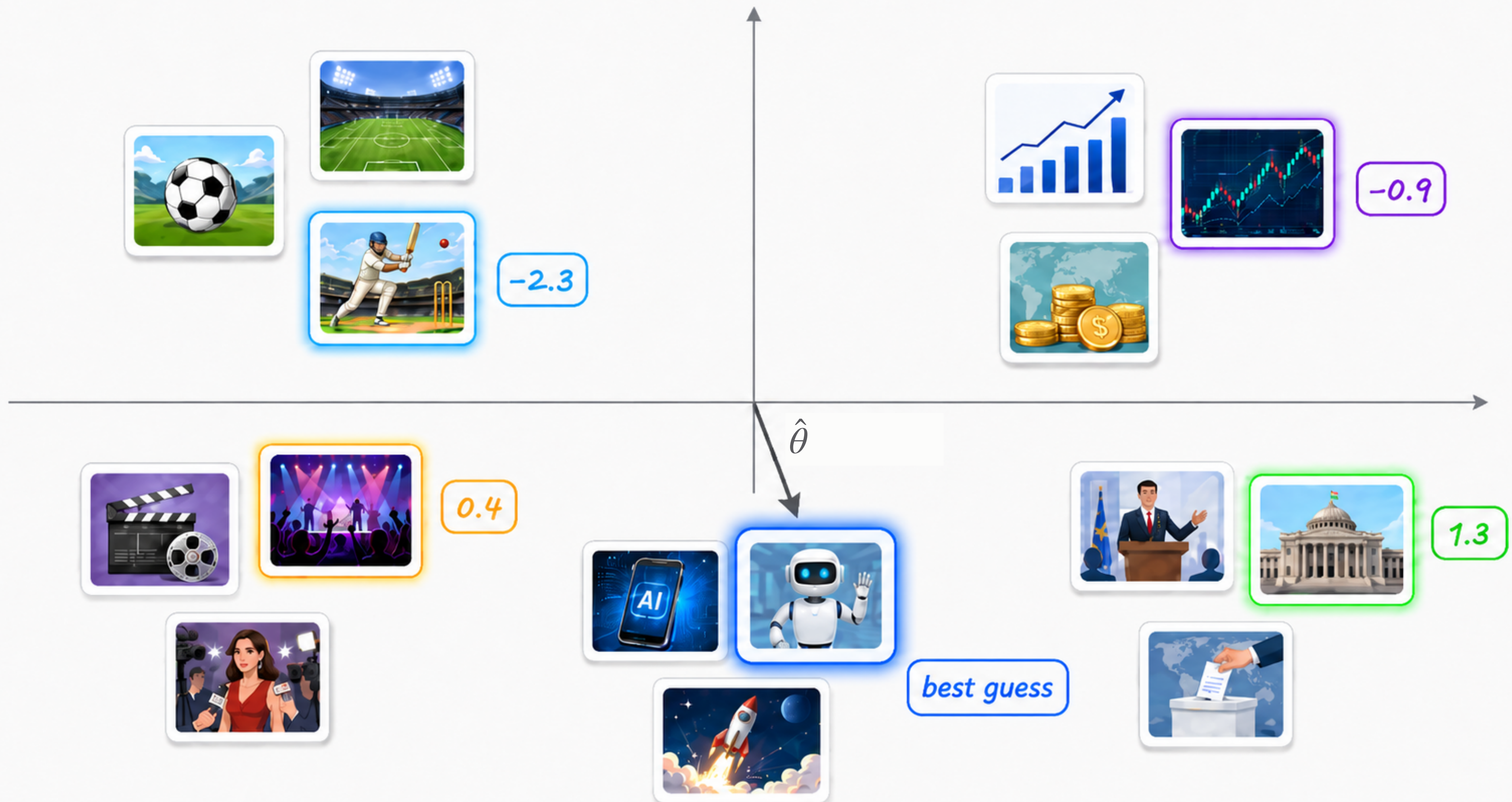
A Worked Example



A Worked Example



A Worked Example



Towards A Bandit Algorithm

Is the best guess enough?

Given the least squares estimate $\hat{\theta}_n$ after n rounds,
what arm do we recommend at time $n + 1$?

- The natural rule: pick the arm with the highest predicted reward

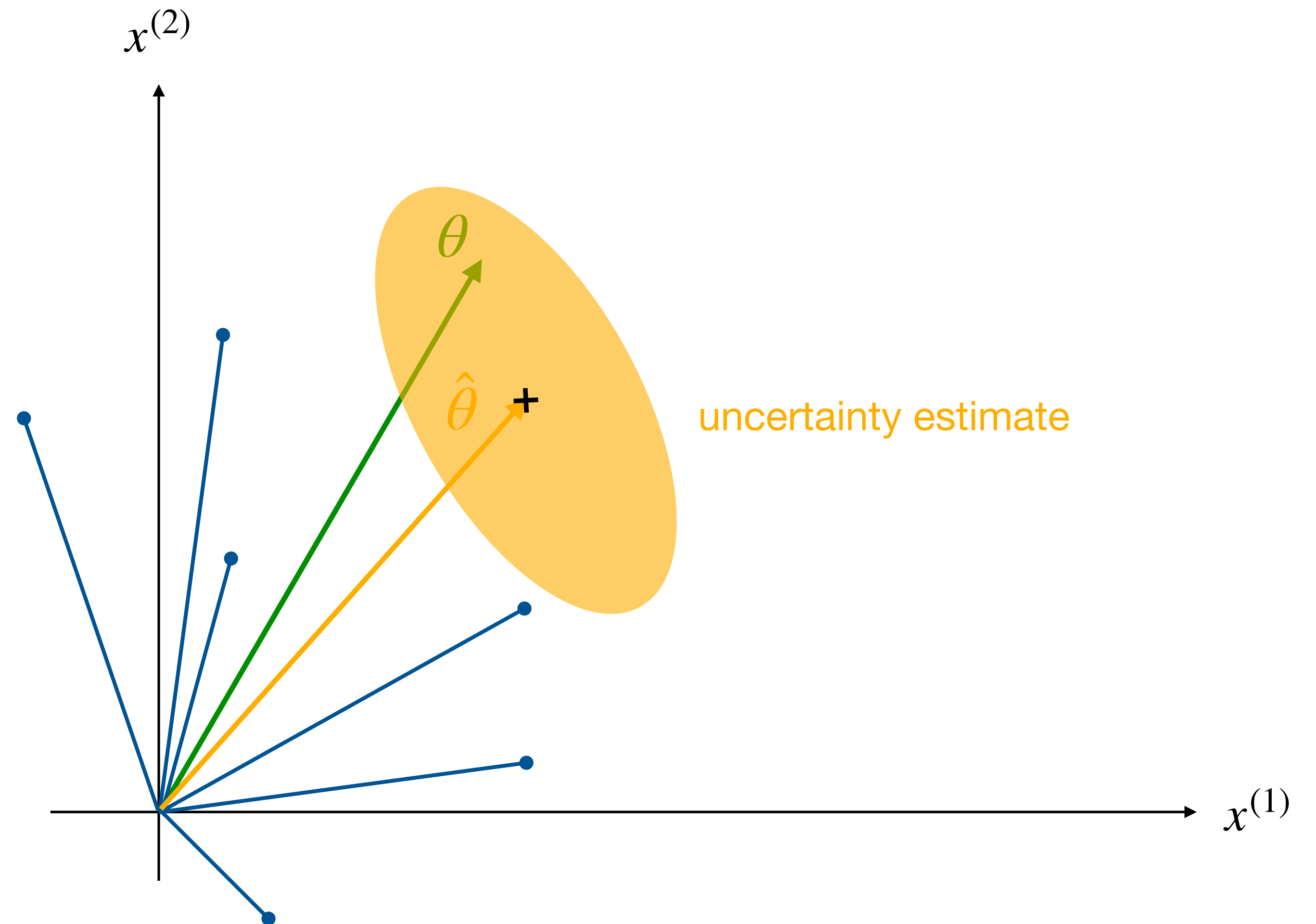
$$a_{n+1} = \arg \max_a \langle \hat{\theta}, x_{t,a} \rangle$$

- Easy to implement—but we know from basic bandits that this won't work!
 - All estimates are imperfect $\Rightarrow$ greedy can get stuck on the wrong arm
 - We can do ε -greedy. But surely there are better approaches?

Intelligent Exploration

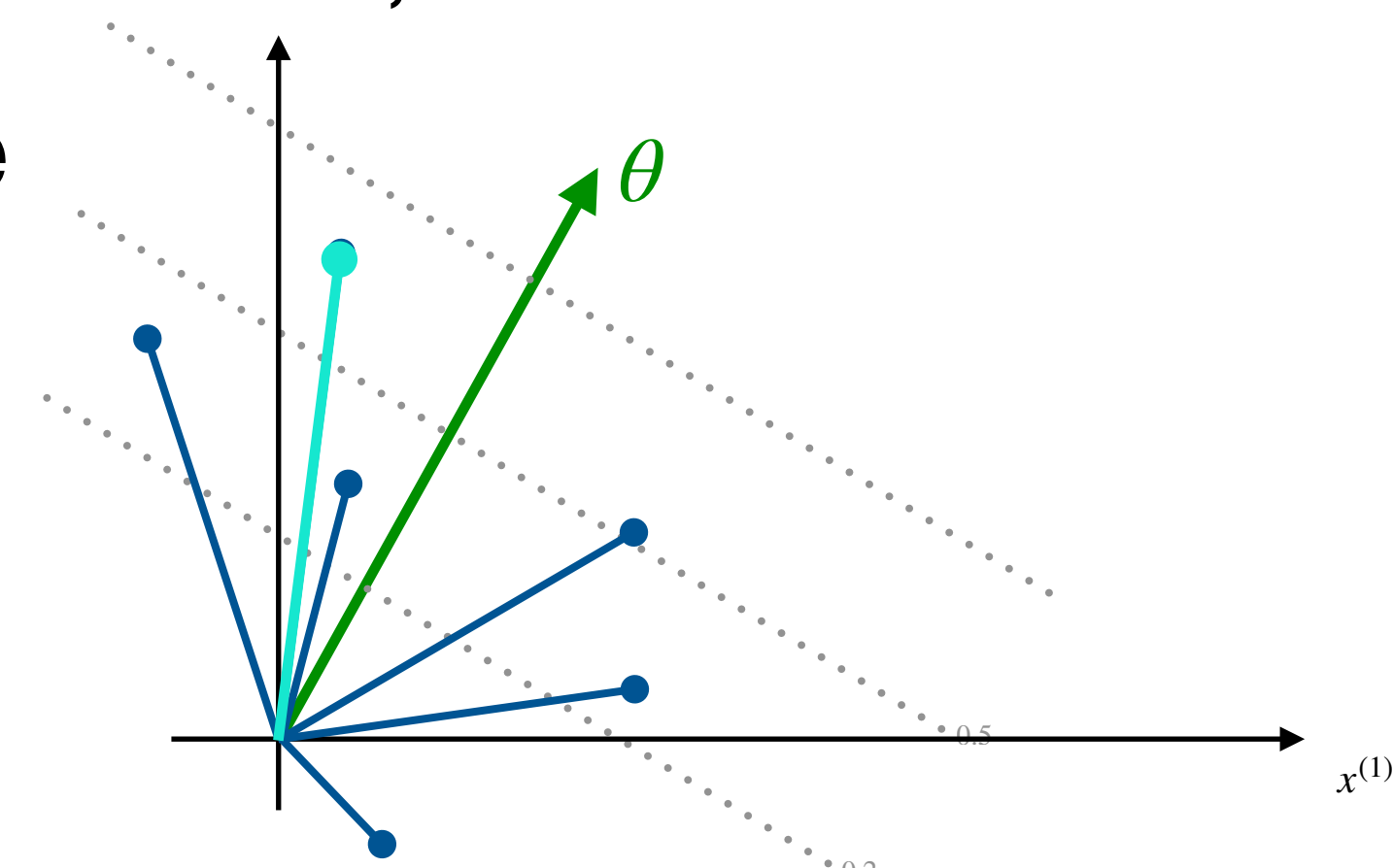
What would it look like?

- To explore intelligently, we revisit concepts we have already seen for bandits: optimism (UCB) or sampling (Thompson).
- Both need one ingredient: *a measure of how uncertain our estimate is*



Conclusion

- We saw the contextual bandit model, where each arm has features
- Features allow us to generalise across users and items—useful in practical systems
- Both user and item features can be combined to handle personalisation and arm churn
- Modelling rewards as a linear function of features gives us linear bandits, which has linear regression (least-squares estimation) at its core
- Next time: capturing uncertainty about $\hat{\theta}$ and designing LinUCB



Homework Logistics

- Homework 1 solution has been posted online
- Homework 2, based on UCB and TS (Lectures 4, 5, 6) will be released on Thursday
 - Due date: Sunday, 14 June, 11:59 PM
 - Similar structure as HW1
- Homework 3, based on linear bandits (Lectures 7, 8, 9): release date TBD
- TAs: Sukanya: da25d900@smail.iitm.ac.in
Roshan: bt21d401@smail.iitm.ac.in

Questions?